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Mass Spectrometry Core Facility

Guidelines

Intact Mass

Proteomics

Crosslinking - MS

CROSSLINKING - MS

XL-MS

Cross-linking mass spectrometry (XL-MS) is a powerful technique to investigate the structure of proteins and protein complexes. As such, it provides (low resolution) distance information between amino acids within the folded protein structure and between interacting proteins by making use of small chemical reagents (most typically NHS-esters).

Sample requirements

- Amine-free buffers (e.g. no Tris or ammonium salts)
- XL-MS can be applied to almost any protein (complex) regardless of size, solubility and strength of interaction.

Crosslinkers

- DSS/BS3
- Zero-length crosslinker: DMTMM, EDC
- Phox (enrichable)
- DSS0 (gas-phase cleavable)

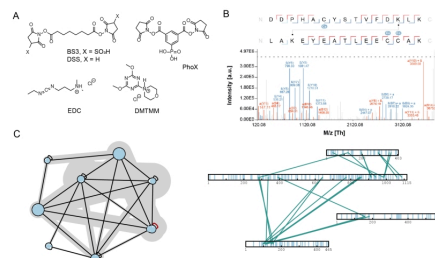


Figure 1 A) Structures of common crosslinking reagents. B) Typical spectrum of an identification of crosslinked peptide with full b- and y-ion series. C) Illustration of interaction networks of protein complexes.

The mass spectrometric detection of crosslinks is not trivial due to sub-stoichiometric reaction efficiencies of only 1-5%. In a direct measurement after the digestion, the sought-for cross-linked peptides are completely overwhelmed by the linear peptides in terms of both numbers and abundance.

Therefore, we offer tailored **state-of-the-art** mass spectrometry workflows: [1,2]

- Offline pre-fractionation of crosslinked peptides by highpH or SEC (for DSS/BS3, DSS0, DMTMM/EDC crosslinked samples)
- Enrichment of crosslinked samples using Fe-IMAC (Phox)
- Phox is a trifunctional cross-linking reagent decorated with a stable phosphonic acid handle. This makes crosslinks amenable to the well-established immobilized metal affinity chromatography (IMAC) enrichment. The handle allows for 300x enrichment efficiency and 97% specificity.

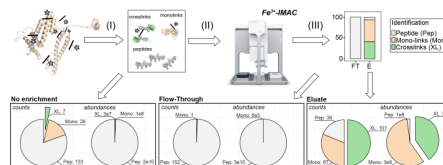


Figure 2 After cross-linking the proteins in their native state, the proteins are denatured, reduced, alkylated, and digested to peptides (I). The mixture of peptides contains 3 different kinds of products: unmodified peptides (gray), monolinks (orange), and cross-links (green) (II). These colors are reused. Cross-linked and monolinked peptides are enriched using Fe-IMAC on a liquid sample handling platform providing high sample throughput (III). Direct measurement of the cross-linked peptides produces low counts of cross-link identifications due to their extremely low abundances (left panel), while measurement following Fe-IMAC enrichment results in no cross-link identifications in the flow-through with similar abundance levels for the linear peptides as detected in the No enrichment (middle panel) and many in the eluate due to their enhanced abundance levels (right panel).

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